DATE : 13.10.2026

DT/NT : DT

LESSON : AWS

SUBJECT : EC2

BATCH : B 418/DE014

AWS-DEVOPS

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techproeducation.com +1 (585) 304 29 59

Table of Contents

- Introduction to EC2
- Instance Types and Pricing
- Security Groups and SSH
- Apache Web Server Hands-On

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Introduction to EC2

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Introduction to EC2

What is EC2?



- EC2 stands for Elastic Compute Cloud in AWS.
- EC2 is a service that allows you to run application programs in the computing environment.
- EC2 is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers.

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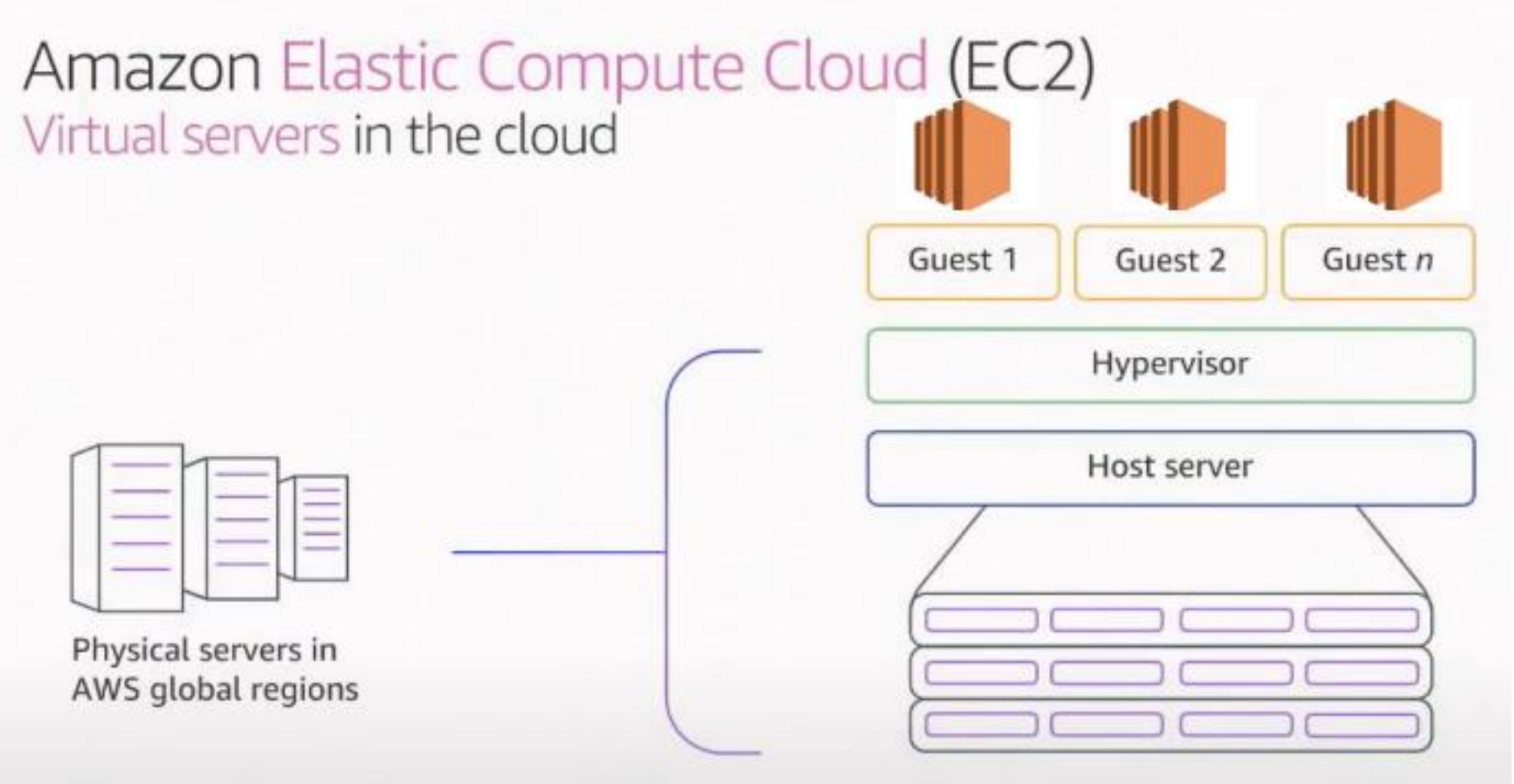
Introduction to EC2

What is EC2?



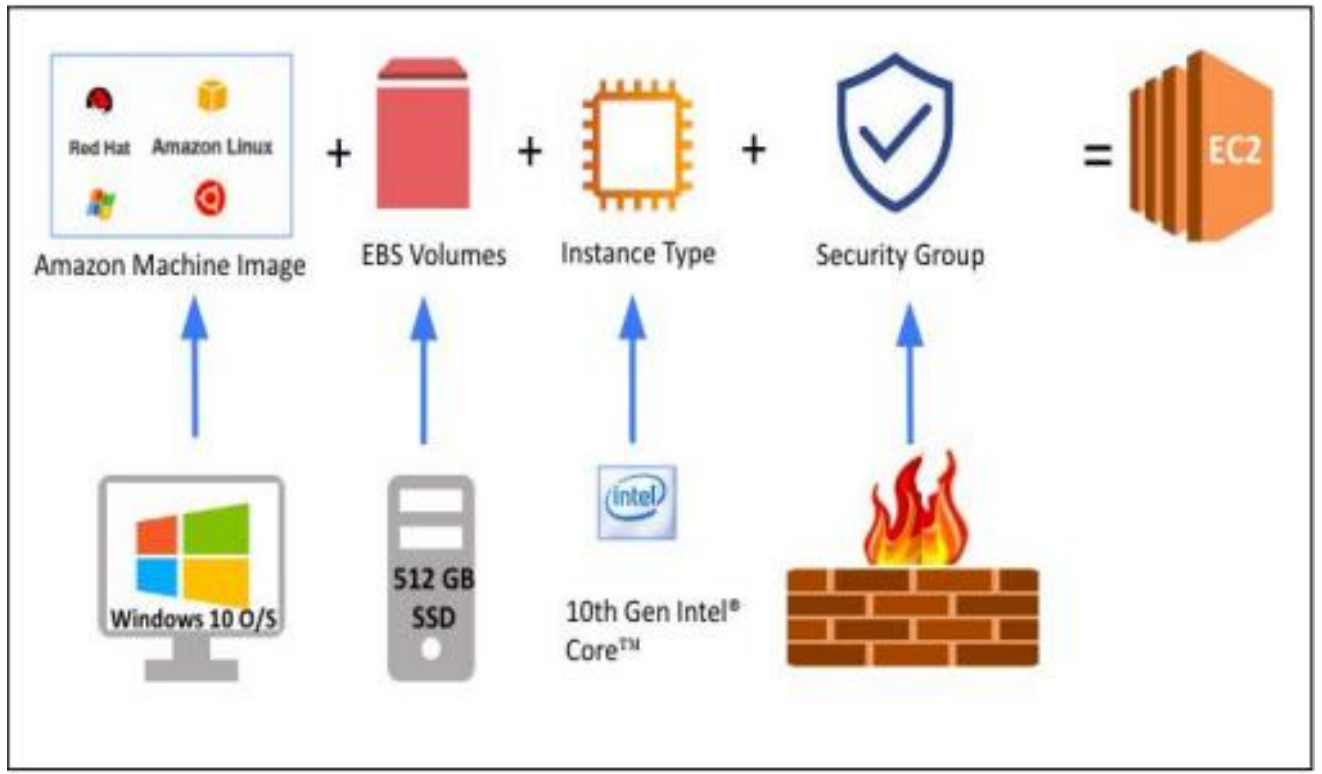
Introduction to EC2

What is EC2?



Introduction to EC2

What is EC2?



- In fact, EC2 is a kind of computer such as your desktop in your home. Components of the EC2 are similar to conventional computer devices.
- Each EC2 component refers to one of the conventional computer parts such as Operation System, Hard Disk and Intel/AMD processors, etc.

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Introduction to EC2

EC2 Features









- Pay only for the resources you use.
- Launch an instance in minutes; time may vary.
- Choose CPU, memory, storage and network capacity.
- Start, stop, resize or terminate instances as needed.

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Introduction to EC2

EC2 Instance Lifecycle

Launch

Create a new EC2 instance

Running

Compute charges generally apply

Stopped

Compute stops; EBS may still cost

Terminated

Instance cannot be restarted

Reboot restarts the OS but does not move the instance to Stopped.

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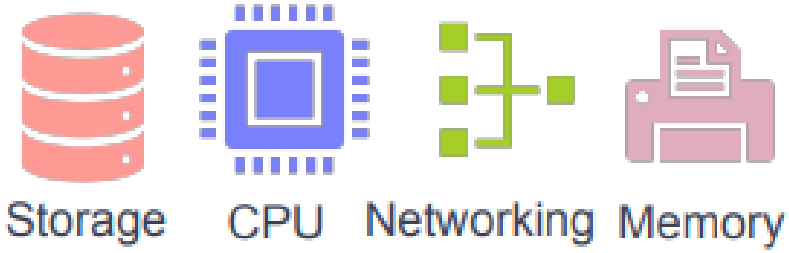
Types of Instances

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EC2 Instances

Types of Instances





- Amazon EC2 provides a wide selection of instance types optimized to fit different use cases.
- Instance types comprise varying combinations of CPU, memory, storage, and networking capacity

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EC2 Instances

AMI vs. Instance Type vs. EBS

AMI

Operating system image and initial software.

Example: Amazon Linux 2023

Instance Type

Computing resources such as vCPU, RAM and networking.

Example: t3.micro

EBS Volume

Persistent block storage for the operating system and data.

Example: 8 GiB

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EC2 Instances

Instance Selection: Two Decisions

1 Instance Type

Select CPU, memory, storage and networking based on the workload.
Example: m7i.large

2 Purchasing Option

Select how to pay or reserve capacity.
Example: On-Demand or Spot.

Choosing an instance type is different from choosing how to pay for it.

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EC2 Instances

EC2 Billing & Purchasing Options

On-Demand

No long-term commitment

Savings Plans

1- or 3-year spend commitment

Reserved Instances

Commit to a matching configuration

Spot Instances

Spare capacity; interruption risk

Dedicated Hosts / Instances

Single-tenant hardware choices

Capacity Reservations

Reserve capacity in a specific AZ

These options serve different goals; they are not six instance families.

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EC2 Instances

On Demand Instances







- Pay for eligible running time without a long-term commitment.
- Per-second billing commonly has a 60-second minimum.
- Scale usage up or down as needed.
- Check rates and related charges for your Region.

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EC2 Instances

On Demand Instances



On-Demand is a good choice for:

- Short-term workloads and development labs.
- Unpredictable traffic that cannot tolerate Spot interruptions.
- Workloads without a long-term usage commitment.

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EC2 Instances

On-Demand Billing: Illustrative Example

Assume an example compute rate of $0.020 per hour.

Usage	Billed compute time	Illustrative compute charge
25 seconds	60 seconds (minimum)	$0.00033
30 minutes	1,800 seconds	$0.01000
24 hours	86,400 seconds	$0.48000

Teaching example only — not a current AWS price quote. Excludes EBS, public IPv4 and other costs.

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EC2 Instances

Reserved Instances (RI)







- Reserved Instances (RIs) can save up to 72% versus On-Demand prices.
- Choose a 1- or 3-year commitment.
- Standard and Convertible RIs offer different flexibility.
- Savings Plans are often preferable when usage may change.

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EC2 Instances

Reserved Instances (RI)



Reserved Instances suit:

- Predictable, steady EC2 usage.
- Teams able to commit for one or three years.
- Workloads that benefit from a matching instance discount.
- A Zonal RI may include an AZ capacity reservation; a Regional RI does not.

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EC2 Instances

On-Demand Capacity Reservations

- Reserve matching EC2 compute capacity in one Availability Zone.
- Help ensure capacity when your instances need to launch.
- You are billed for unused reserved capacity.
- A Capacity Reservation does not automatically provide a discount.

Replaces obsolete Scheduled Reserved Instances

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EC2 Instances

Example: Reserve Capacity for a Sales Event

1 Plan

An online store expects a short period of unusually high demand.

2 Reserve

Reserve matching EC2 capacity in the required Availability Zone.

3 Launch

Launch compatible instances into the reservation when needed.

Capacity reservations address availability of compute capacity, not high availability by themselves.

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EC2 Instances

Spot Instances

Discounted spare EC2 capacity

- Spot Instances use spare EC2 capacity at a discount.
- Discounts may reach up to 90% versus On-Demand.
- AWS can reclaim capacity and interrupt an instance.
- Choose Spot for workloads able to tolerate interruptions.

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EC2 Instances

Spot Instance



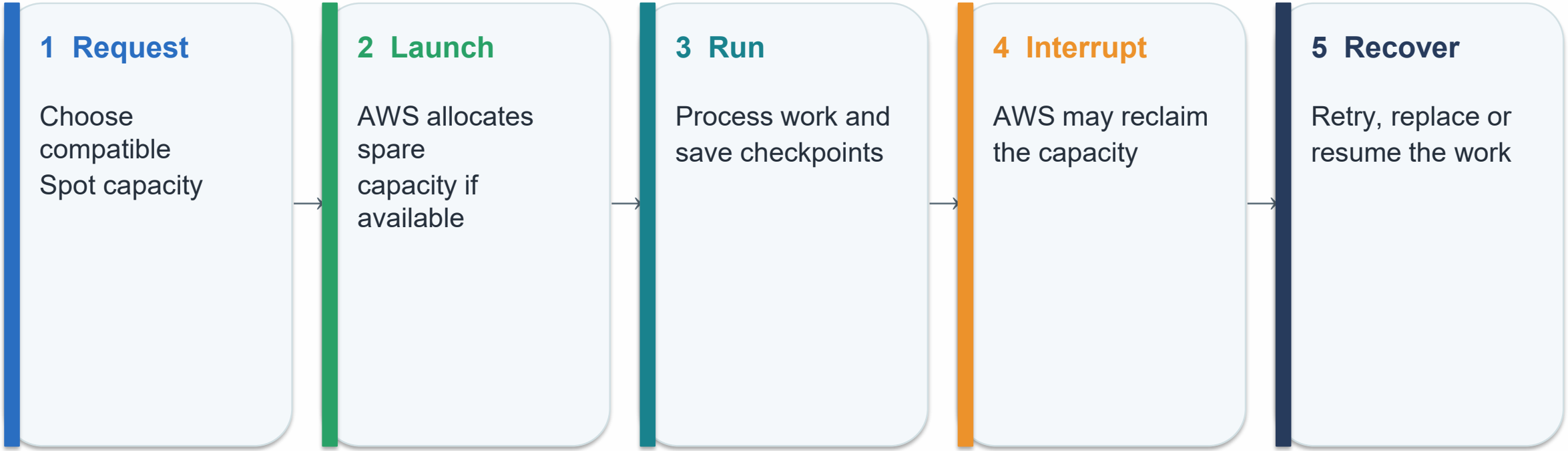
Spot Instances work well for:

- Batch processing and data analysis.
- Automated testing and temporary build workers.
- Fault-tolerant background jobs.
- Work that can pause, retry or run on replacement instances.

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EC2 Instances

How Spot Instances Work



Spot interruption notices are best-effort; applications should be designed to recover safely.

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EC2 Instances

Spot vs. On-Demand: Key Trade-offs

On-Demand

- No Spot reclamation interruptions
- Predictable per-unit rate
- Suitable for important steady services
- Still not guaranteed to be outage-free

Spot

- Lower prices when capacity is available
- Interruptions may occur
- Ideal for batch and fault-tolerant jobs
- Not suitable for a fragile single server

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EC2 Instances

Dedicated Host/Instance



- Dedicated Host: an entire physical server dedicated to one customer, with host-level visibility and supported license options.
- Dedicated Instance: instance on single-tenant hardware, but without the same host-level control.
- Consider these for licensing or compliance requirements.

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EC2 Instances

Savings Plans

Commit to consistent usage (USD/hour) for 1 or 3 years to reduce eligible compute costs.

Compute Savings Plans

Flexible across eligible EC2, AWS Lambda and AWS Fargate usage.

EC2 Instance Savings Plans

Discount eligible usage within a chosen instance family and AWS Region.

Savings and flexibility depend on the plan. Check AWS Pricing before committing.

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EC2 Instances

Pricing Decision Guide

Short or uncertain workload

On-Demand

Interruptible batch jobs

Spot

Predictable long-term use

Savings Plans / RIs

Reserve capacity in an AZ

Capacity Reservation

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EC2 Instances

Purpose: Choose the Right Resources

What the application needs

CPU-heavy? Choose compute optimized.
Large data in RAM? Choose memory optimized.

How the EC2 instance is paid for

On-Demand? Spot? Savings Plans?
Purchasing choice is a separate decision.

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EC2 Instances

Main EC2 Instance Categories

General Purpose

M, T — balanced usage

Compute Optimized

C — CPU-intensive

Memory Optimized

R, X, U — large memory

Storage Optimized

I, D — local storage I/O

Accelerated Computing

G, P, Inf, Trn — accelerators

High Performance Computing

Hpc — simulations, HPC jobs

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EC2 Instances

General Purpose

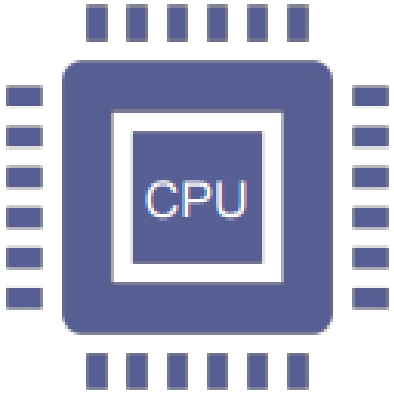


- Balance CPU, memory and networking for many applications.
- Good for web servers, small databases and app servers.
- Examples: M family (balanced), T family (burstable).
- Burstable T instances use CPU credits.

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EC2 Instances

Compute Optimized



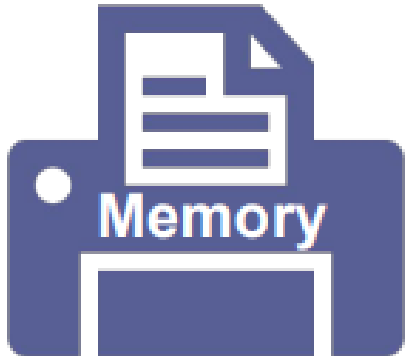
- Compute Optimized instances are ideal for compute bound applications that benefit from high performance processors.

- Instances belonging to this family are well suited for batch processing workloads, media transcoding, high performance web servers, dedicated gaming server, etc.

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EC2 Instances

Memory Optimized



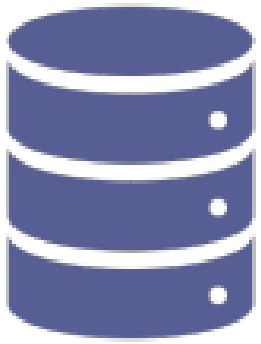
• Memory optimized instances are used in situations requiring a high-performance database, real-time large data analytics, and high memory usage.

• There are R, X, Z and U type instances in this category.

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EC2 Instances

Storage Optimized



- Designed for high-performance local storage workloads.
- Useful for NoSQL systems, data processing and fast I/O.
- Examples include I-family and D-family instances.
- Local instance-store data is not a replacement for durable EBS or backups.

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EC2 Instances

Accelerated Computing



- Accelerators (GPUs or specialized chips) speed up certain tasks.
- Used for graphics, ML training and ML inference.
- Example families: G and P (GPUs), Inf (inference), Trn (training).
- Choose only if the workload benefits from an accelerator.

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EC2 Instances

Reading an EC2 Instance Name

r7i.4xlarge

- r = Memory-optimized instance series.
- 7 = Generation (not necessarily the newest).
- i = Intel processor option.
- 4xlarge = Instance size.
- Check which generations are currently available.

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EC2 Instances

EC2 Instance Name Anatomy



Family	Generation	Option	Size
Memory optimized	Generation 7	Intel processors	Instance size

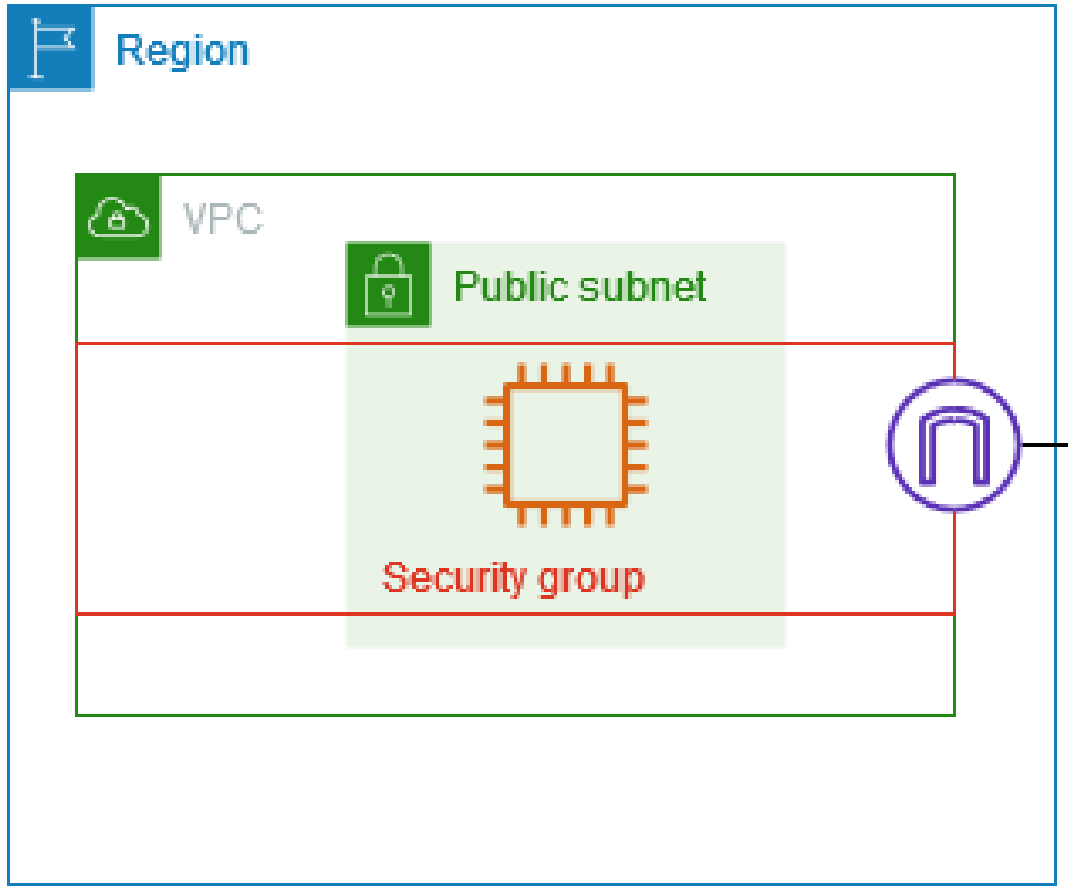
Family + generation + option + size = an EC2 instance type

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Security Groups

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Security Groups

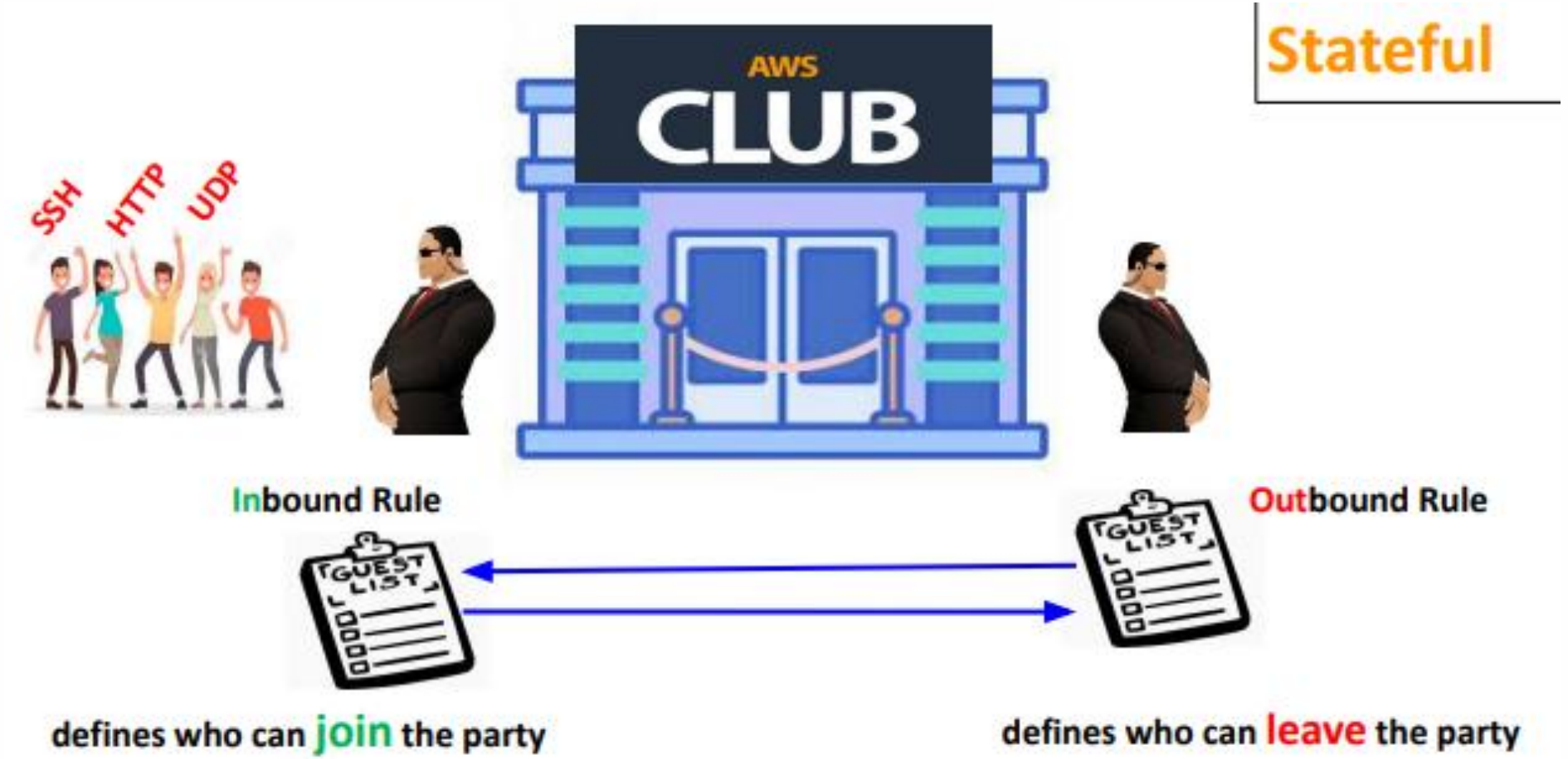




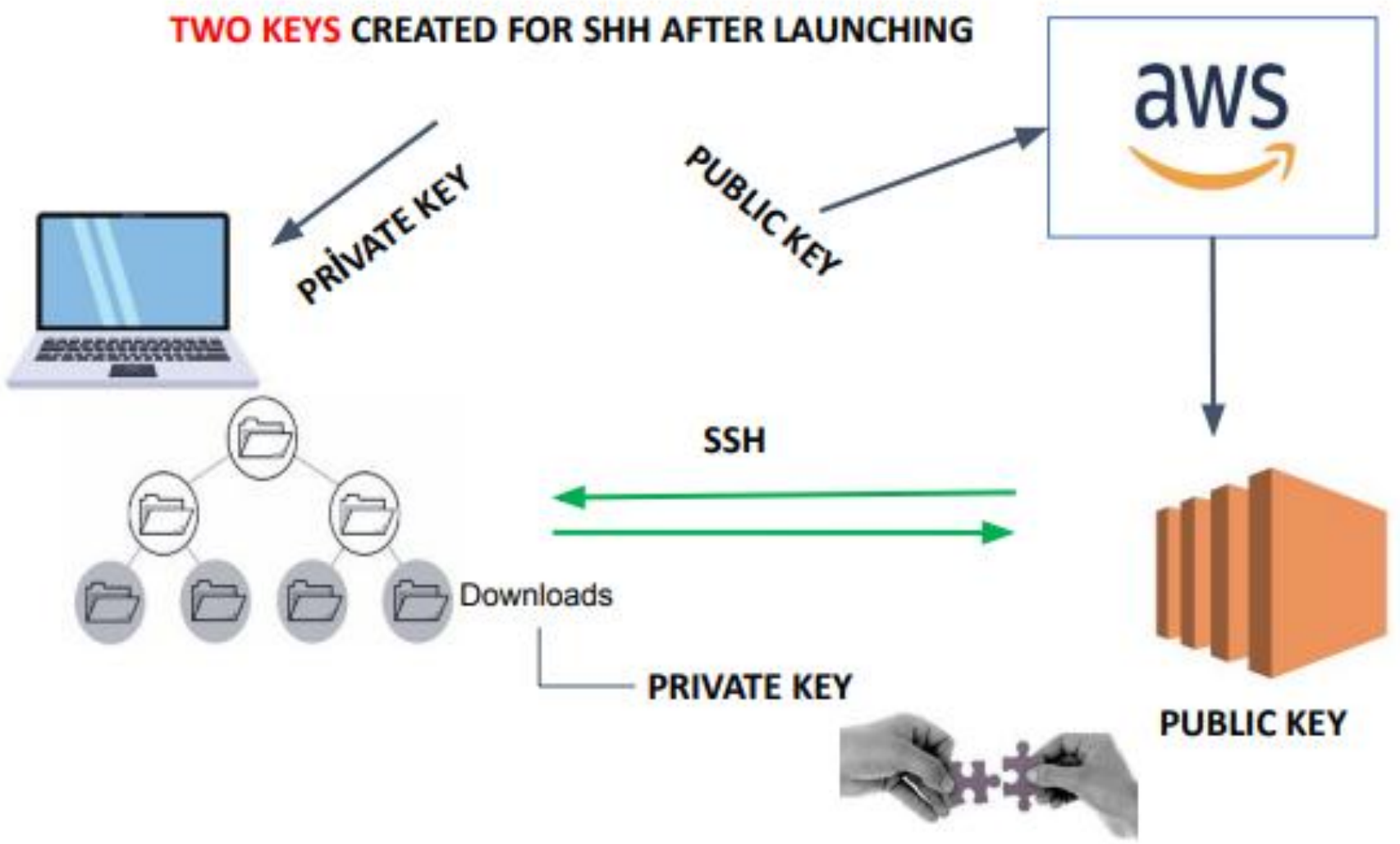
Security groups:

- Allow rules only
- Inbound + outbound
- Stateful filtering
- No explicit deny

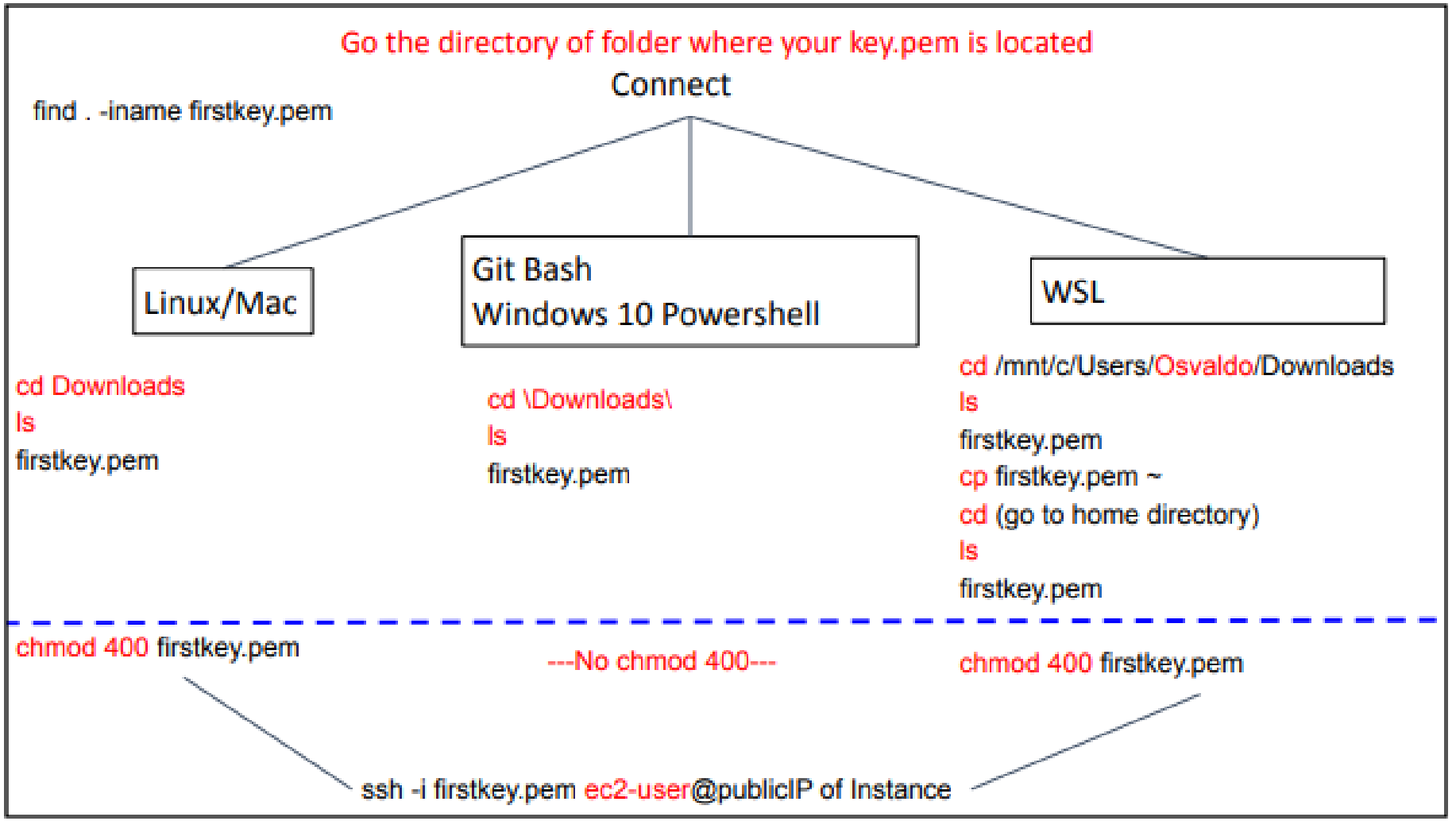
Security Groups



Recap


TWO KEYS CREATED FOR SHH AFTER LAUNCHING
aws
PRİVATE KEY
PUBLIC KEY
SSH
Downloads
PRIVATE KEY
PUBLIC KEY

Recap


Go the directory of folder where your key.pem is located
Connect
find . -iname firstkey.pem
Linux/Mac
Git Bash
Windows 10 Powershell
WSL
cd Downloads
ls
firstkey.pem
cd \Downloads\
ls
firstkey.pem
cd /mnt/c/Users/Osvaldo/Downloads
ls
firstkey.pem
cp firstkey.pem ~
cd (go to home directory)
ls
firstkey.pem
chmod 400 firstkey.pem
---No chmod 400---
chmod 400 firstkey.pem
ssh -i firstkey.pem ec2-user@publicIP of Instance

Recap



soil = ?

APPLICATION

EC2

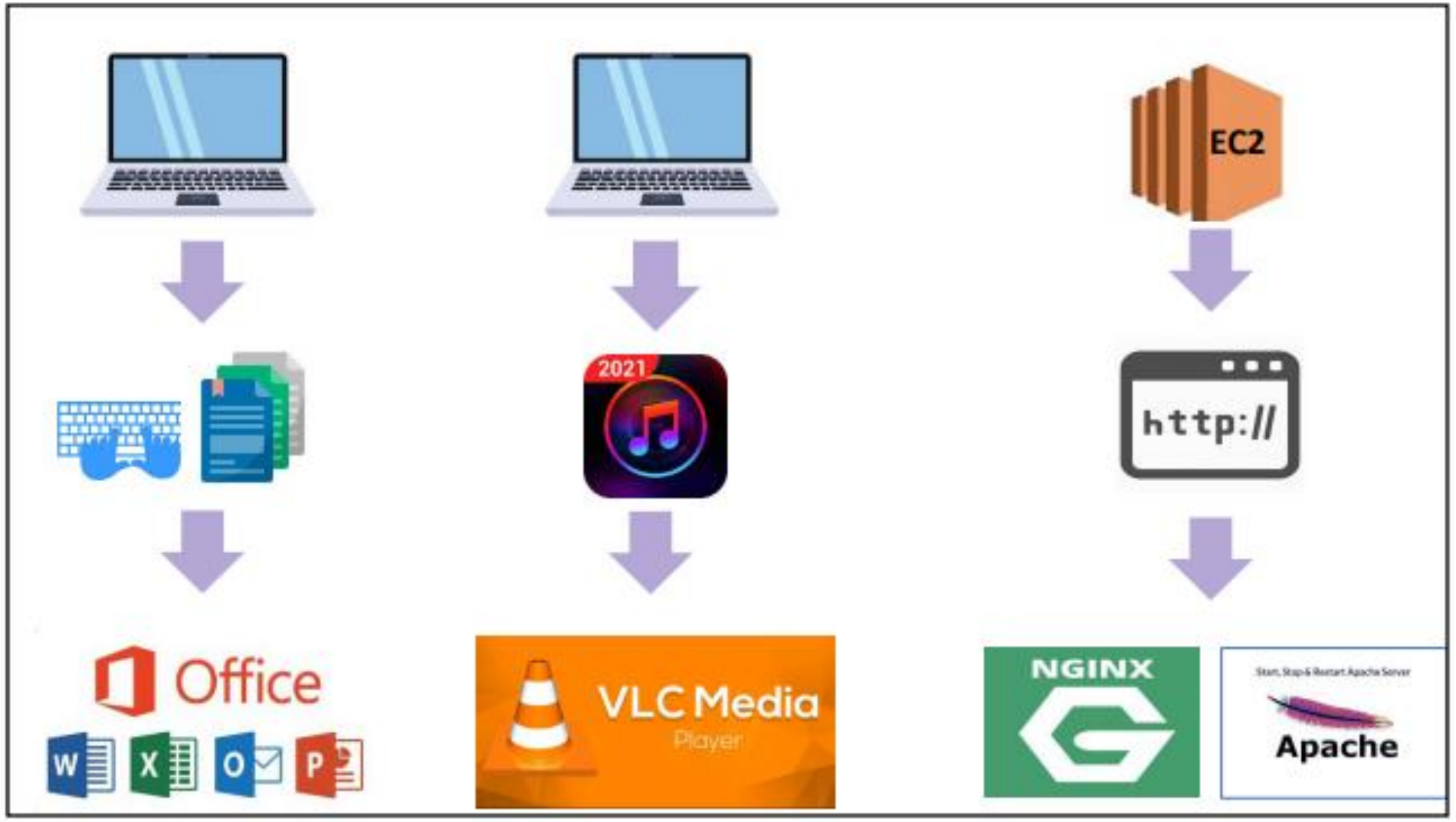




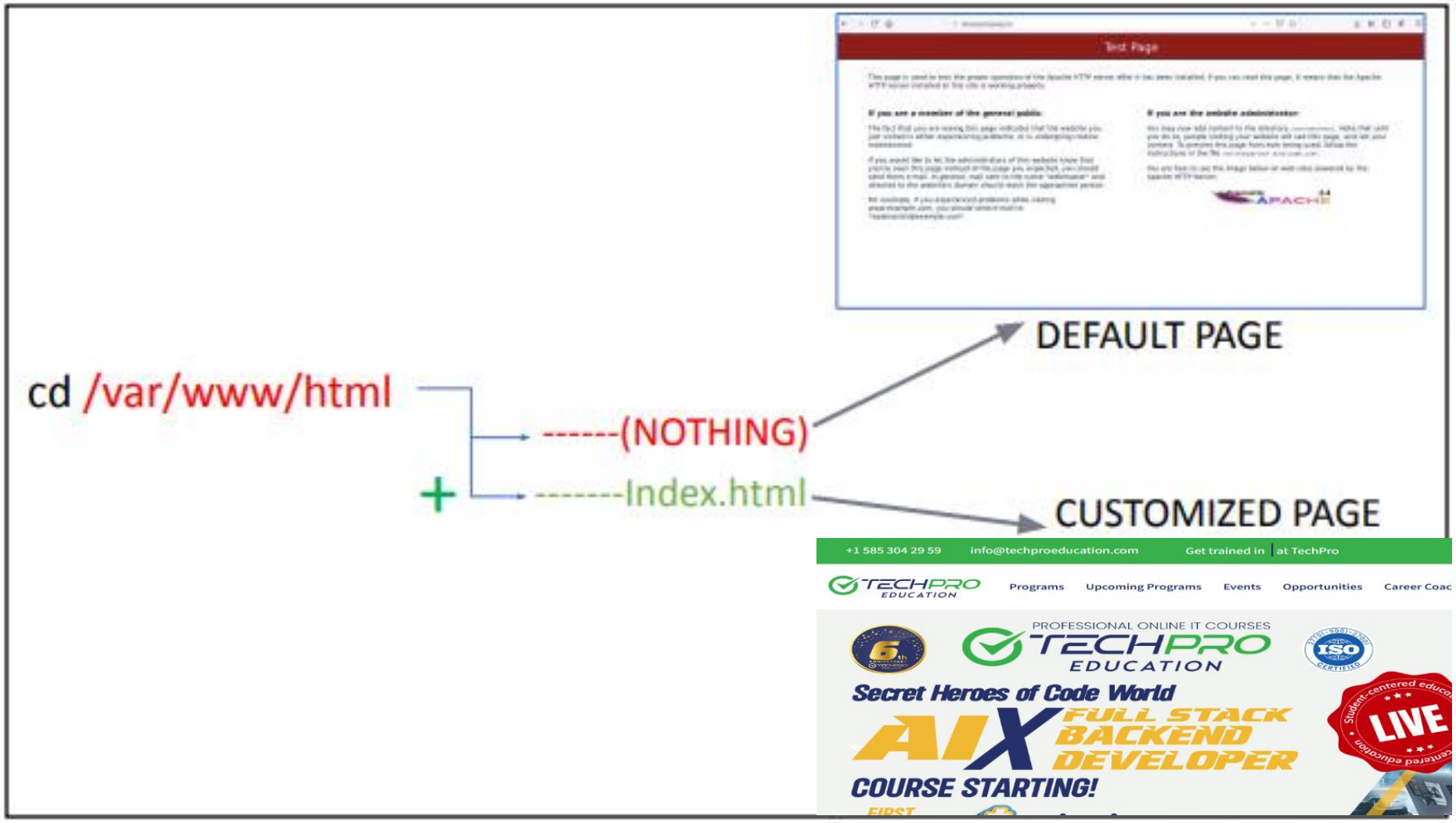


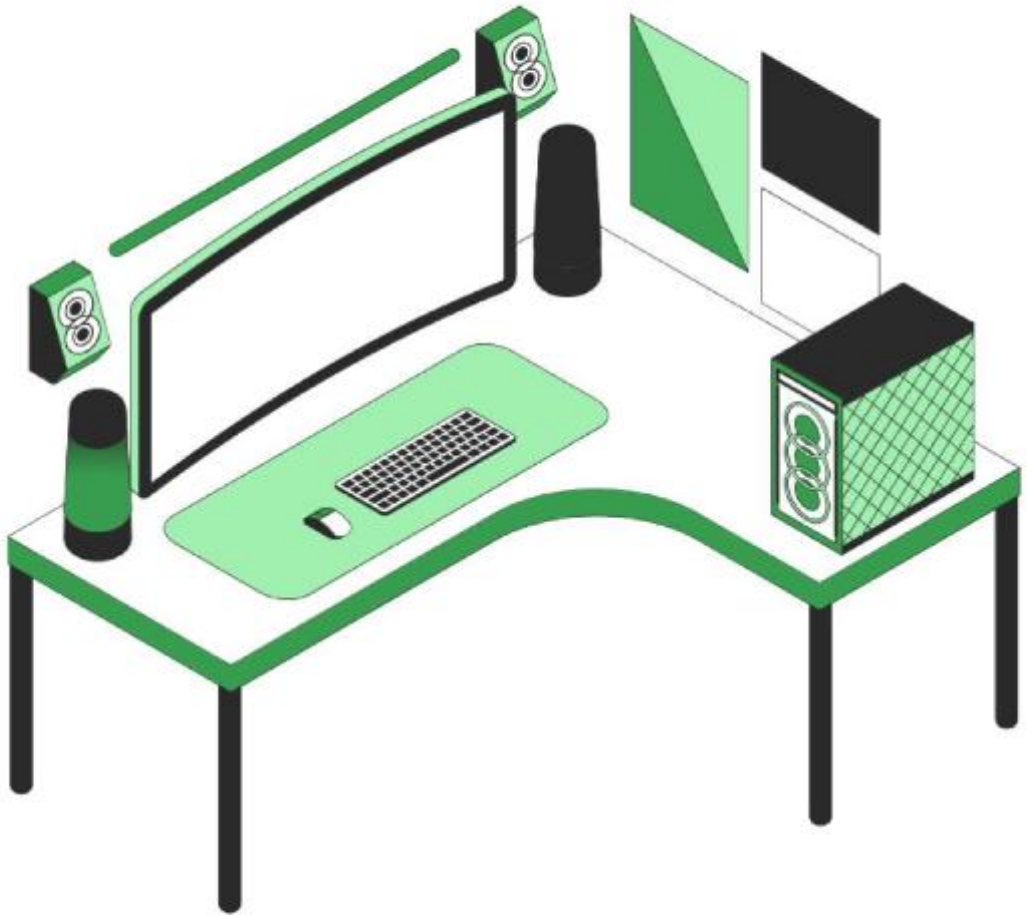
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Recap



Recap


cd /var/www/html
------(NOTHING)
+
-------Index.html
DEFAULT PAGE
CUSTOMIZED PAGE
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info@techproeducation.com
Get trained in
at TechPro
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Upcoming Programs
Events
Opportunities
PROFESSIONAL ONLINE IT COURSES
Secret Heroes of Code World
AIX
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BACKEND
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COURSE STARTING!
LIVE



Do you have any questions?

Send it to us! We hope you learned something new.

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